



3D Combustion Simulation of Non Pre-mixed Liquid Spray Combustion

Spring 2026

Title: 3D Combustion Simulation of Non Pre-mixed Liquid Spray Combustion.

Presented by:

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Name of the course: Theory of combustion and Emissions.

Course code: ME60232

Course Instructor: Prof. Mahendra Reddy.

Name of the department: Mechanical Engineering Department.

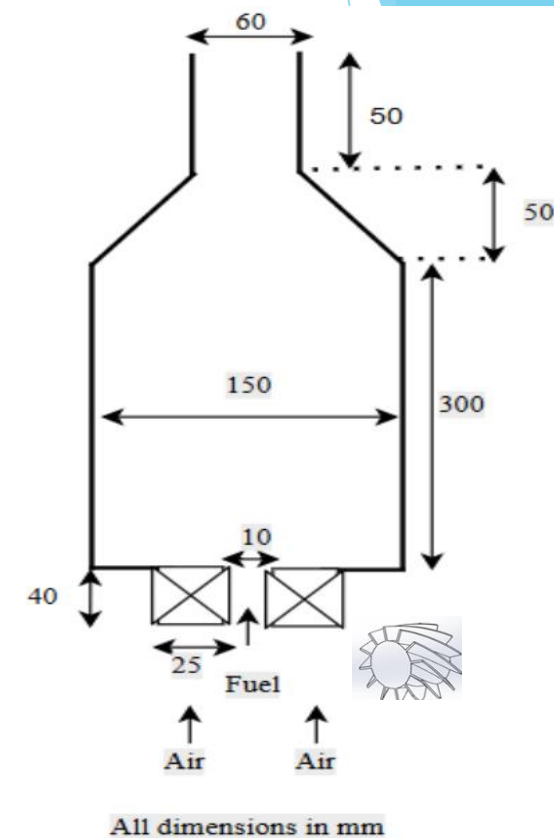
Name of the Institute: Indian Institute of Technology Kharagpur.

Problem Statement

- The objective of this study is to perform a 3D CFD simulation of liquid fuel combustion in a swirl-stabilized combustor using ANSYS Fluent. The geometry of the combustor is provided and includes a 60° swirler through which air enters, generating a strong recirculation zone for flame stabilization.
- The working fluid consists of air as the oxidizer and kerosene as the liquid fuel.
- The fuel is injected as a spray with a droplet diameter of 13 μm and a spray cone angle of 45°.
- The Energy input rate of the fuel is 30kW to be obtained using of equivalence ratios ($\phi = 0.8, 1.0, 1.2$). Both fuel and oxidizer enter the domain at an initial temperature of 300 K.

The results to be found from the simulation are:

- ☐ To obtain contours of static temperature and velocity fields.
- ☐ To evaluate mass fractions of key species: OH, kerosene vapor, CO, and NO_x species (NO, NO₂).
- ☐ To plot variation of exit temperature and NO_x emissions with equivalence ratio ϕ .



Geometry of the swirl combustor

For Geometry -

Swirl wall Thickness – 2.5 mm

Vanes Thickness – 2 mm

No of Vanes – 12 mm

Combustor wall thickness – 5 mm

Introduction

- Combustion is a rapid exothermic chemical reaction between a fuel and an oxidizer, resulting in the release of heat, light, and reaction products.
- In most engineering systems, it involves hydrocarbon fuels reacting with oxygen to produce energy.

Essential Requirements for Combustion

- **Fuel:** Substance that releases energy (e.g., kerosene)
- **Oxidizer:** Usually air (oxygen) to support reaction
- **Mixing Mechanism:** Proper contact between fuel and oxidizer
- **Ignition Source:** Initial energy to start the reaction
- **Combustion Chamber:** Confined space to sustain and control combustion
- **Sufficient Residence Time:** Allows complete reaction and heat release

Combustion type (Based on state of matter of fuel):

- **Gaseous Combustion:** Fuel and oxidizer mix at molecular level (e.g., natural gas burners).
- **Liquid Spray Combustion:** Liquid fuel atomized into droplets, evaporates and burns (e.g., gas turbines, aircraft engines — this study).
- **Solid Fuel Combustion:** Combustion of solid materials like coal or biomass involving devolatilization and char burning.

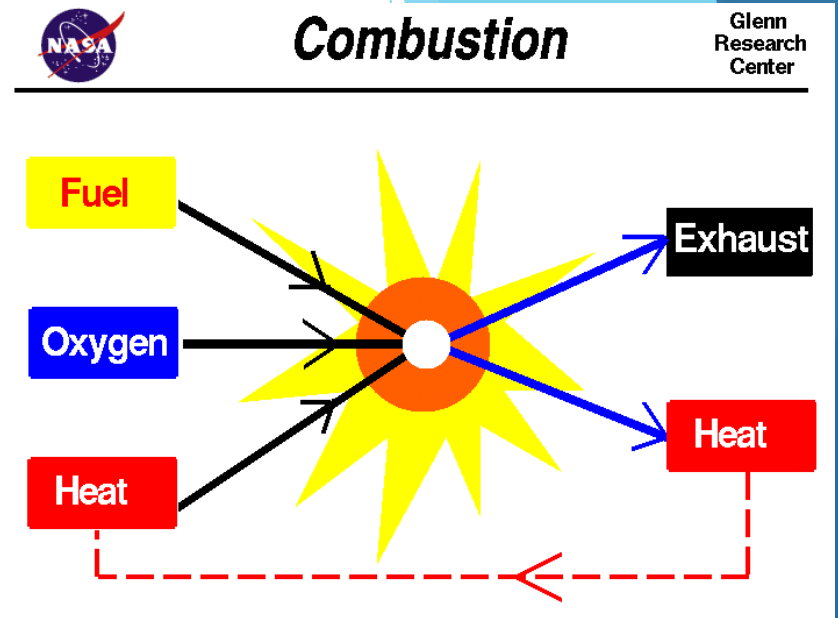


Image Courtesy : Glenn Research centre, NASA

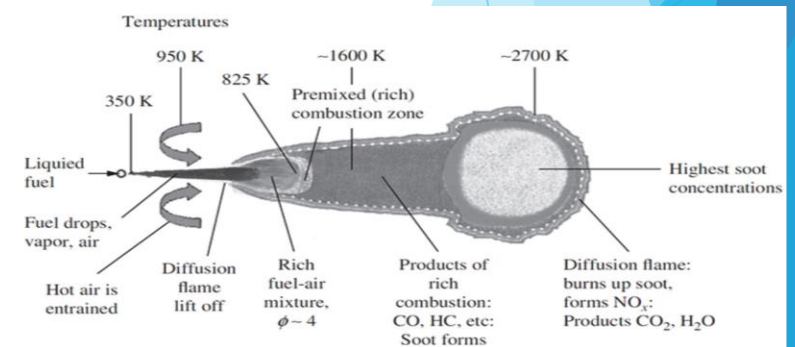


Image Courtesy : Lecture slides – Prof. Pabhitra Bahduk – Dept of Mech Engg IITKGP.

Introduction (continuation)

Liquid Spray Combustion & Discrete Phase Modelling (DPM)

- Liquid spray combustion involves the injection of liquid fuel into a hot, oxidizing environment where it undergoes atomization, evaporation, mixing, and chemical reaction. Unlike gaseous combustion, the fuel must first transition from liquid to vapor before burning, making it a multi-physics process.
- **Role of Swirl in Spray Combustion** – Creates recirculation zone enhancing mixing and evaporation that stabilizes flame thus improving combustion efficiency .

Key Physical Processes

- ❑ **Fuel into Injector → Atomization → Droplet Transportation → Evaporation → Mixing with oxidiser → Combustion → Heat & Products.**

Discrete Phase Model (DPM) in CFD:

- Treats liquid fuel as discrete particles (droplets) in a continuous fluid (air)
- Tracks droplet trajectories using Lagrangian approach
- Supports two-way coupling (interaction between droplets and airflow)
- Accurately captures spray physics
- Models droplet evaporation → fuel vapor formation
- Flame structure
- Temperature distribution
- Emissions (CO, NO_x, OH)

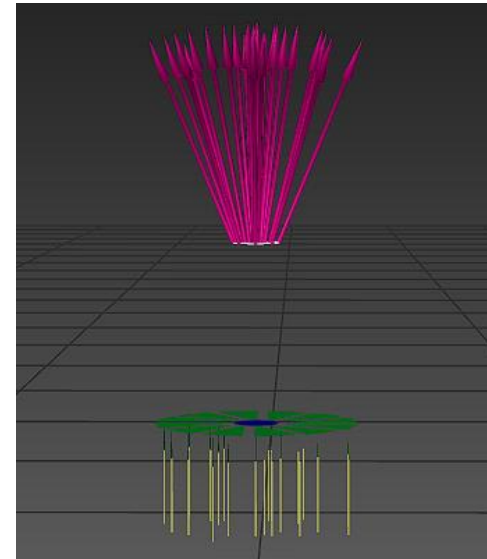


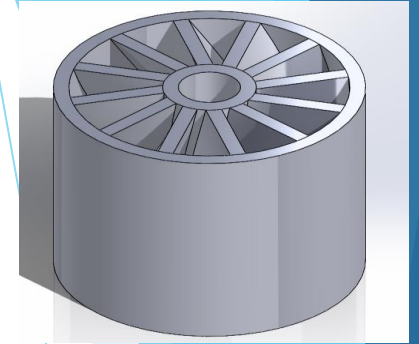
Image Courtesy :
display of injection of
fluid droplets – ANSYS
fluent .

Methodology



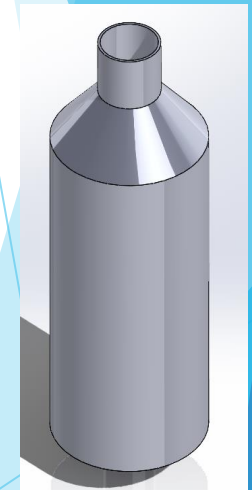
3D Model designing phase

- Designed 60° swirler vanes in SolidWorks .
- Assembled it with 3D combustion chamber geometric model in SolidWorks.
- Defined boundary faces (air inlet, fuel inlet, wall, outlet) in Ansys SpaceClaim.
- Assigned named selections and exported geometry for meshing.



Pre-processing phase

- ANSYS Fluent – 3D , 4 solvers with double precision.
- Imported mesh into ANSYS Fluent (.msh file).
- Displayed and verified boundary faces.



Solver Setup

- Pressure based Solver with Absolute velocity formulation and steady time – General Setup.
- k-ε realizable turbulence model with scalable near wall functions for low computation requirements.

Phase 3
Cold flow
(air only)

Air-only boundary conditions
Mass flow inlet, outlet + walls

Initialize flow field
Solution → Initialization → Hybrid

Run cold flow simulation
Run → Calculate; watch residuals

Converged

Species transport model
Non-premixed PDF or EDC model

Discrete phase model (DPM)
13 μm droplet, 45° spray cone

NOx post-processing model
Thermal + prompt + N₂O pathways

Initialize from cold flow
From converged cold flow state

Patch temperature: 1500–2000 K
In combustion / recirculation zone

Not
converged

Cold flow run

- Applied air-only boundary conditions (mass flow inlet).
- Performed hybrid initialization to establish initial flow field.
- Ran cold flow simulation until convergence to obtain stable velocity and recirculation .

Combustion Model Setup

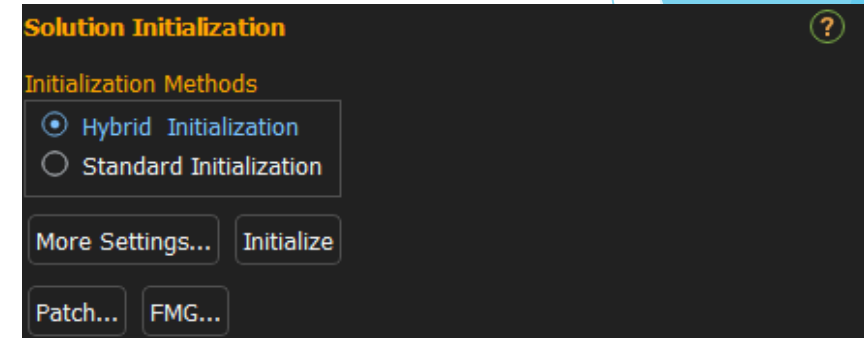
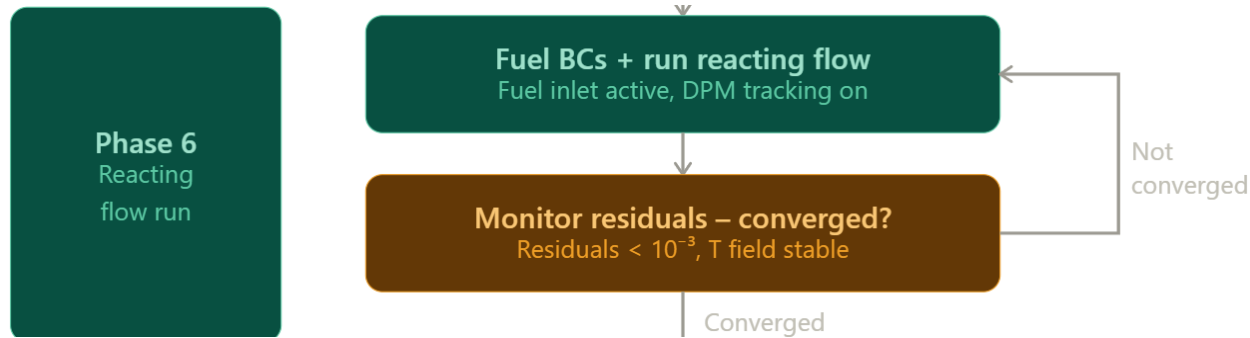
- Enabled non-premixed combustion model (PDF approach) for reacting flow.
- Configured Discrete Phase Model (DPM) with 13 μm droplets and 45° spray cone.
- Activated NOx model (thermal and prompt mechanisms) for emission prediction (post 1000 iterations).

Patch and combustion

- Initialized solution using converged cold flow field.
- Applied temperature patch in recirculation zone to initiate ignition (by creating a cell register at Injection site).
- Started reactive simulation to achieve flame stabilization and combustion.

Phase 4
Combustion
model setup

Phase 5
Patch &
ignite



Reacting flow run

- Activated fuel injection with DPM tracking and enabled combustion model.
- Performed reactive flow simulation with proper boundary conditions.
- Monitored residuals (target $< 10^{-3}$) and ensured temperature field stabilization.
- Completed around **40,000 iterations** to achieve convergence.



Post processing

- Extracted species mass fractions (OH, kerosene vapor, CO, NO, N₂O).
- Generated contour plots of static temperature and velocity fields.
- Plotted exit temperature and NO_x variation with equivalence ratio at outlet using Area-Weighted Average .

Calculations of flow rates for different Equivalence ratios

- Given data - Heat input $Q = 30 \text{ kW}$, Equivalence ratios $\phi = 0.8, 1.0, 1.2$.
- Gathered data – LCV of kerosene = 43134 kJ/kg [1] , Stoic A/F ratio = 15.6 [1].

We know that $Q = \dot{m}_f * \text{LCV}$, From the equation $\dot{m}_f = 0.000696 \text{ kg/s}$.

Equivalence ratios $\phi = \frac{\left[\frac{A}{F}\right]_{\text{Stoic}}}{\left[\frac{A}{F}\right]_{\text{Actual}}} \rightarrow \left[\frac{A}{F}\right]_{\text{Actual}} = \frac{\left[\frac{A}{F}\right]_{\text{Stoic}}}{\phi} \rightarrow \dot{m}_{\text{air}} = \dot{m}_f * \frac{\left[\frac{A}{F}\right]_{\text{Stoic}}}{\phi}$

□ For $\phi = 0.8$, $\dot{m}_f = 0.000696 \text{ kg/s}$, Stoic A/F ratio = 15.6

$\dot{m}_{\text{air}} = 0.013562 \text{ kg/s}$, Actual A/F ratio = 19.5

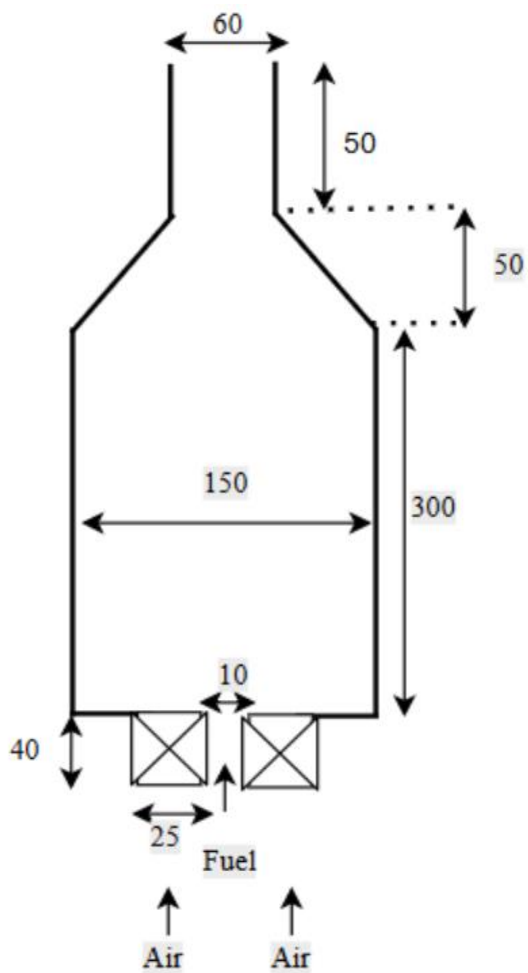
□ For $\phi = 1.0$, $\dot{m}_f = 0.000696 \text{ kg/s}$, Stoic A/F ratio = 15.6

$\dot{m}_{\text{air}} = 0.010850 \text{ kg/s}$, Actual A/F ratio = 15.6

□ For $\phi = 1.2$, $\dot{m}_f = 0.000696 \text{ kg/s}$, Stoic A/F ratio = 15.6

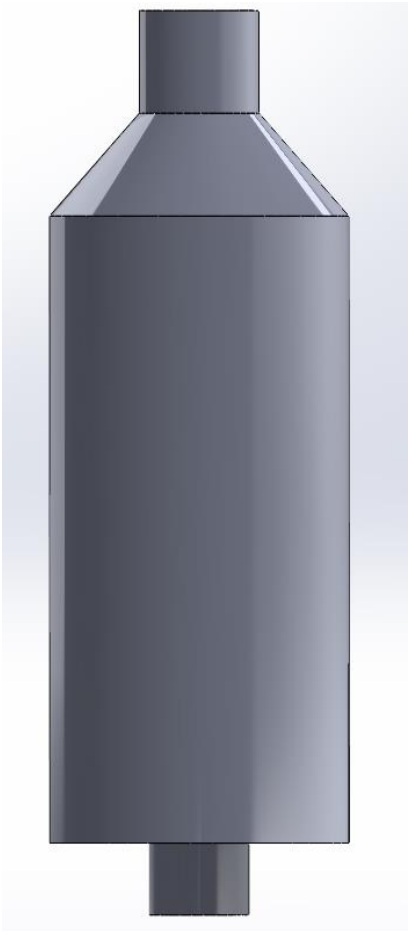
$\dot{m}_{\text{air}} = 0.009042 \text{ kg/s}$, Actual A/F ratio = 13

3D geometric model of combustor from Solidworks

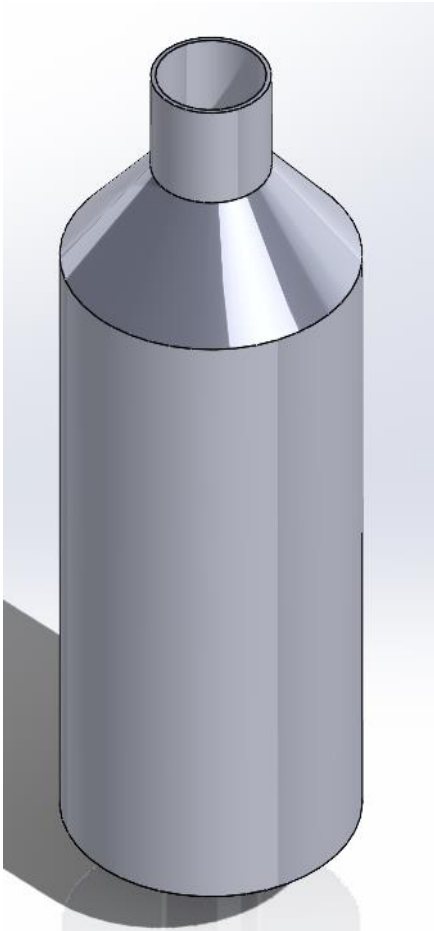


All dimensions in mm

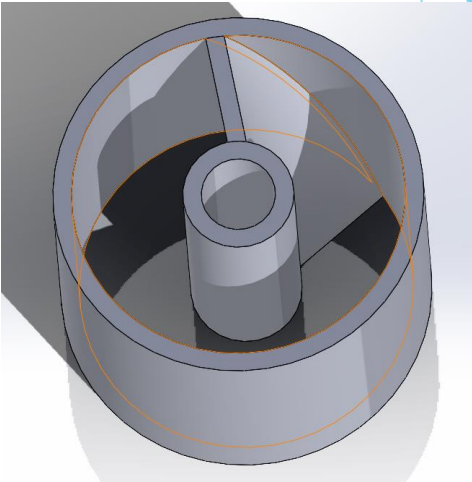
Problem statement dimensions.



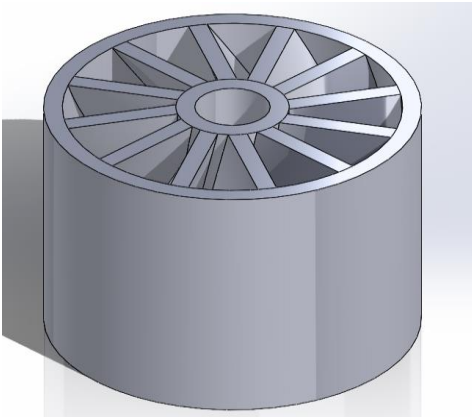
Front view of 3d model



Isometric view of 3d model



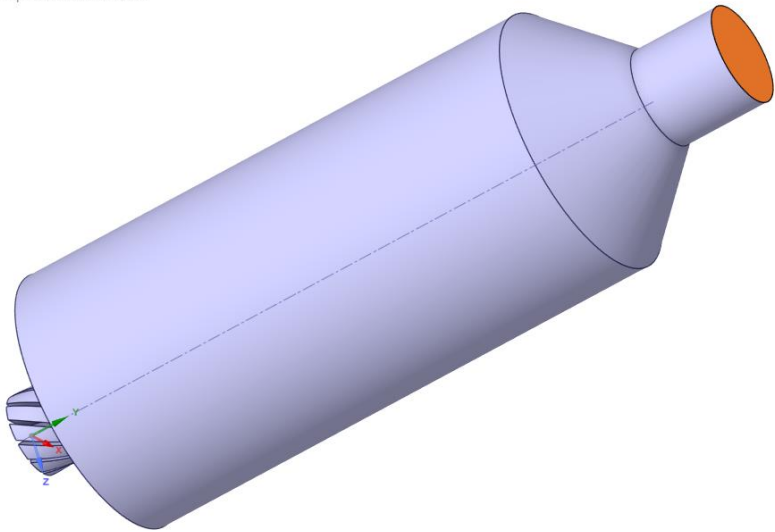
Isometric view of 60° vane angle of a single vane.



Isometric view of Swirler .

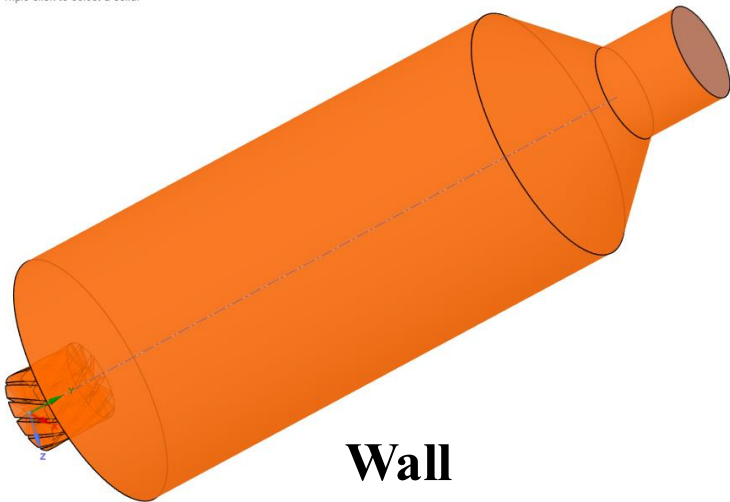
Assigning of boundary faces in ANSYS SpaceClaim.

edge loop. Triple-click to select a solid.

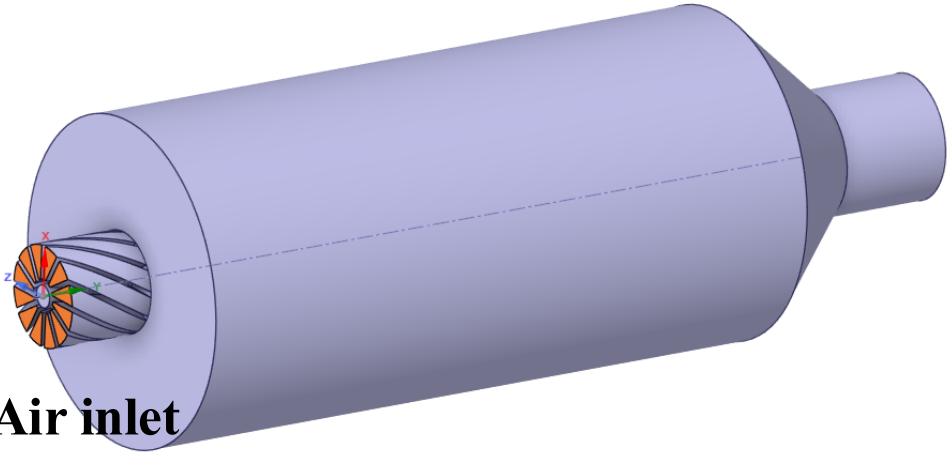


**Outlet
face**

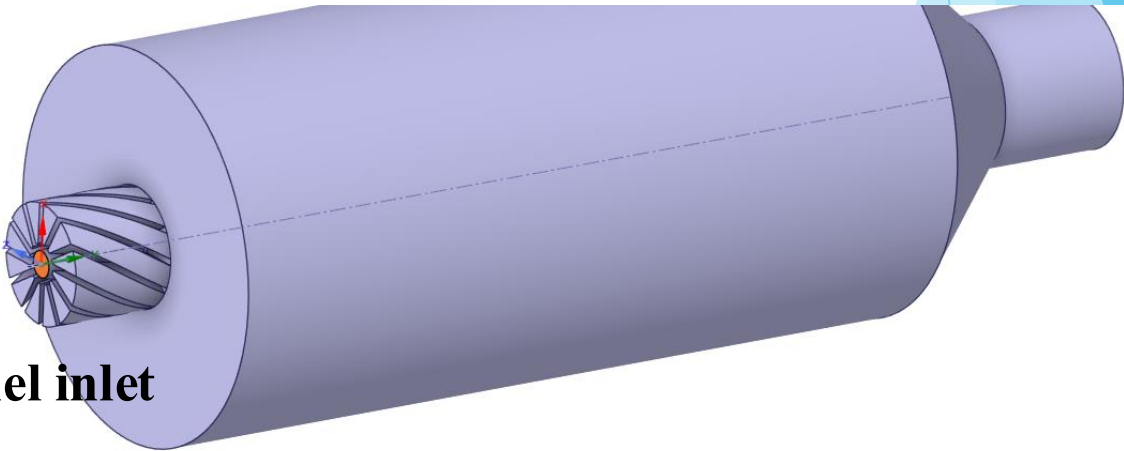
edge loop. Triple-click to select a solid.



Wall



Air inlet



Fuel inlet

Simulation Setup

1. Mesh size and Mesh Quality

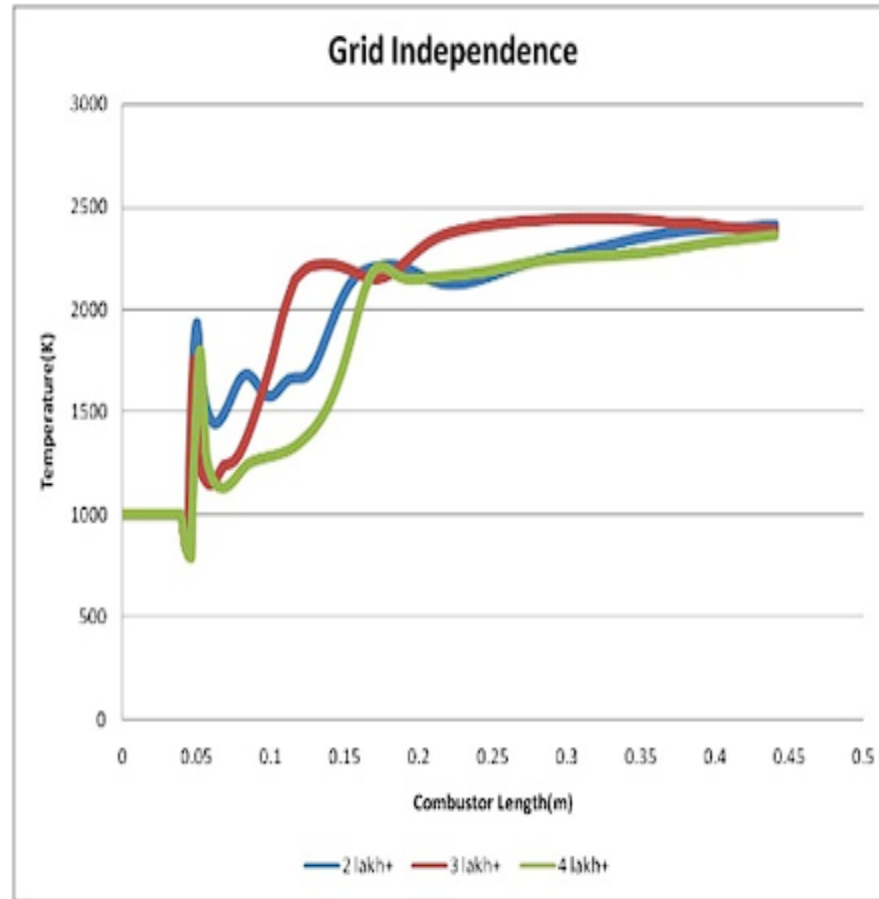
Physics setup

1. Models
2. Material properties
3. Cell Zone conditions
4. Boundary conditions
5. Injection settings - DPM
6. Reference values

Solver Settings

Solution status

- Temperature rises sharply near the inlet(at 0.04m) due to combustion initiation
- Fluctuations are observed in the initial region (0–0.15 m) due to strong mixing
- Beyond 0.2 m, temperature profiles for all meshes almost coincide
- Indicates grid-independent behaviour in the developed region
- 4 lakh mesh gives slightly smoother variation, but overall trend remains the same



Graph: Grid independence: temperature profiles converge beyond 0.2 m (~2–4 lakh elements)

*****Taken reference from group performing the same liquid fuel simulation.**

Mesh Size

Cells	Faces	Nodes
189184	793176	438546

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
volume-volume	Mixed Cell	0.11683253	43.64047

Model	Settings
Space	3D
Time	Steady
Viscous	Realizable k-epsilon turbulence model
Wall Treatment	Scalable Wall Function
Heat Transfer	Enabled
Species	Non-Premixed Combustion
Coupled Dispersed Phase	Enabled
NOx Pollutants	Enabled


Viscous Model

×

Model

☐ Inviscid
☐ Laminar
☐ Spalart-Allmaras (1 eqn)
☒ k-epsilon (2 eqn)
☐ k-omega (2 eqn)
☐ Transition k-kl-omega (3 eqn)
☐ Transition SST (4 eqn)
☐ Reynolds Stress (7 eqn)
☐ Scale-Adaptive Simulation (SAS)
☐ Detached Eddy Simulation (DES)
☐ Large Eddy Simulation (LES)

Model Constants

C2-Epsilon

1.9

TKE Prandtl Number

1

TDR Prandtl Number

1.2

Energy Prandtl Number

0.85

Wall Prandtl Number

0.85

k-epsilon Model

☐ Standard
☐ RNG
☒ Realizable

Near-Wall Treatment

☐ Standard Wall Functions
☒ Scalable Wall Functions
☐ Non-Equilibrium Wall Functions
☐ Enhanced Wall Treatment
☐ Menter-Lechner
☐ User-Defined Wall Functions

Options

☐ Viscous Work
☐ Curvature Correction
☐ Production Limiter

User-Defined Functions

Turbulent Viscosity

none

Prandtl and Schmidt Numbers

TKE Prandtl Number

none

TDR Prandtl Number

none

Energy Prandtl Number

none

Wall Prandtl Number

none

Turbulent Schmidt Number

OK

Cancel

Help

Material Properties

- Solid	
- aluminum	
Density	2719 kg/m ³
Cp (Specific Heat)	871 J/(kg K)
Thermal Conductivity	202.4 W/(m K)
- Mixture	
- pdf-mixture	
Mixture Species	names
Density	pdf
Cp (Specific Heat)	mixing law
Thermal Conductivity	0.0454 W/(m K)
Viscosity	1.72e-05 kg/(m s)

- Droplet Particle	
- kerosene-liquid	
Density	780 kg/m ³
Cp (Specific Heat)	2090 J/(kg K)
Latent Heat	226000 J/kg
Vaporization Temperature	341 K
Boiling Point	477 K
Volatile Component Fraction	100 %
Binary Diffusivity	3.79e-06 m ² /s
Diffusivity Reference Pressure	none
Saturation Vapor Pressure	piecewise linear
Heat of Pyrolysis	0
Temperature Averaging Coefficient	none
Composition Averaging Coefficient	none
Vaporization/Boiling Model	diffusion controlled
Thermolysis Model	none

Boundary Conditions

- Inlet	
- air_inlet	
Reference Frame	Absolute
Mass Flow Specification Method	Mass Flow Rate
Mass Flow Rate [kg/s]	0.013562
Total Temperature [K]	300
Initial Gauge Pressure [Pa]	0
Direction Specification Method	Normal to Boundary
Turbulence Specification Method	Intensity and Hydraulic diameter
Turbulent Intensity [%]	5
Hydraulic diameter [mm]	30
Pollutant NO Mass Fraction	0
Pollutant N2O Mass Fraction	0
Mean Mixture Fraction	0
Mixture Fraction Variance	0
Discrete Phase BC Type	escape

Mass-Flow Inlet

Zone Name: air_inlet

Momentum Thermal Radiation Species DPM Multiphase Potential Structure UDS

Reference Frame: Absolute

Mass Flow Specification Method: Mass Flow Rate

Mass Flow Rate [kg/s]: 0.013562

Supersonic/Initial Gauge Pressure [Pa]: 0

Direction Specification Method: Normal to Boundary

Turbulence

Specification Method: Intensity and Hydraulic Diameter

Turbulent Intensity [%]: 5

Hydraulic Diameter [mm]: 30

Apply Close Help

- Outlet	
- outlet	
Backflow Reference Frame	Absolute
Gauge Pressure [Pa]	0
Pressure Profile Multiplier	1
Backflow Total Temperature [K]	300
Backflow Direction Specification Method	Normal to Boundary
Turbulence Specification Method	Intensity and Hydraulic diameter
Backflow Turbulent Intensity [%]	5
Backflow Hydraulic diameter [mm]	30
Mean Mixture Fraction	0
Mixture Fraction Variance	0
Backflow Pollutant NO Mass Fraction	0
Backflow Pollutant N2O Mass Fraction	0
Discrete Phase BC Type	escape
Backflow Pressure Specification	Total Pressure
Build artificial walls to prevent reverse flow?	no
Radial Equilibrium Pressure Distribution	no
Average Pressure Specification?	no
Specify targeted mass flow rate	no


Pressure Outlet
✕

Zone Name
outlet

Momentum Thermal Radiation Species DPM Multiphase Potential Structure UDS

Backflow Reference Frame Absolute

Gauge Pressure [Pa] 0

Pressure Profile Multiplier 1

Backflow Direction Specification Method Normal to Boundary

Backflow Pressure Specification Total Pressure

☐ Prevent Reverse Flow
☐ Radial Equilibrium Pressure Distribution
☐ Average Pressure Specification
☐ Target Mass Flow Rate

Turbulence
Specification Method Intensity and Hydraulic Diameter
Backflow Turbulent Intensity [%] 5
Backflow Hydraulic Diameter [mm] 30

Apply Close Help

Wall	
Wall Thickness [mm]	5
Heat Generation Rate [W/m^3]	0
Material Name	Aluminum
Thermal BC Type	Heat Flux
Heat Flux [W/m^2]	0
Enable shell conduction?	no
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Roughness Height [mm]	0
Wall Roughness Constant	0.5
Discrete Phase BC Type	trap
Convective Augmentation Factor	1

Wall Motion

☒ Stationary Wall
 ☐ Moving Wall

Motion

☒ Relative to Adjacent Cell Zone

Shear Condition

☒ No Slip
 ☐ Specified Shear
 ☐ Specularity Coefficient
 ☐ Marangoni Stress

Zone Name

Adjacent Cell Zone

Momentum

Thermal

Radiation

Species

DPM

Multiphase

UDS

Potential

Structure

Ablation

Thermal Conditions

☒ Heat Flux
 ☐ Temperature
 ☐ Convection
 ☐ Radiation
 ☐ Mixed
 ☐ via System Coupling
 ☐ via Mapped Interface

Heat Flux [W/m²]

▼

Wall Thickness [mm]

▼

Heat Generation Rate [W/m³]

▼

☐ Shell Conduction

1 Conduction Layer

Edit...

Material Name

▼

Edit...

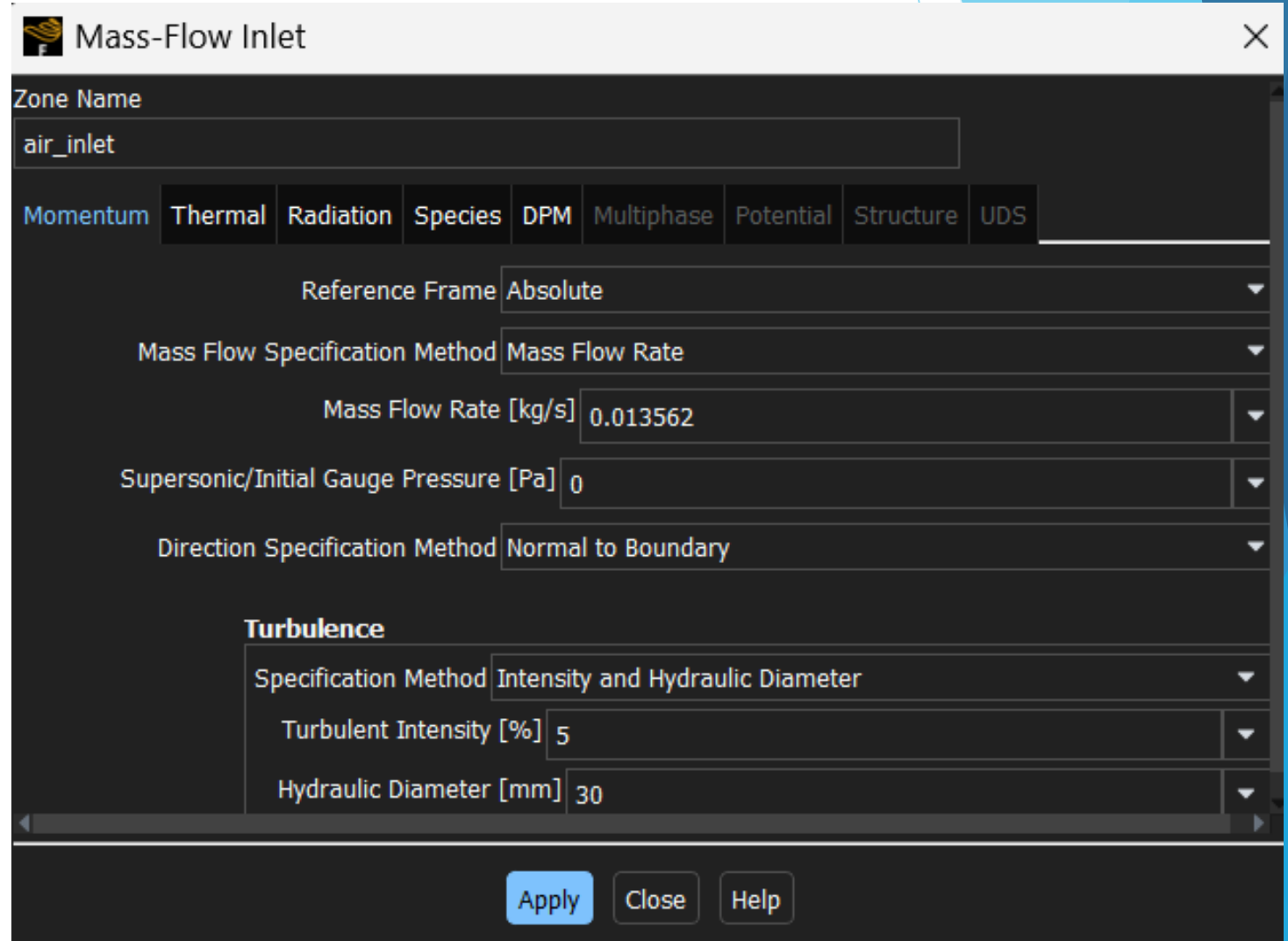
Apply

Close

Help

Cold Flow conditions

- For Cold flow run we just allow only the oxidiser to flow over the geometry by giving the mass flow rate in the In-let Boundary condition.
- Then Initialise and run the calculation.
- Hybrid Initialization is used .
- Run it for 2000 iterations.



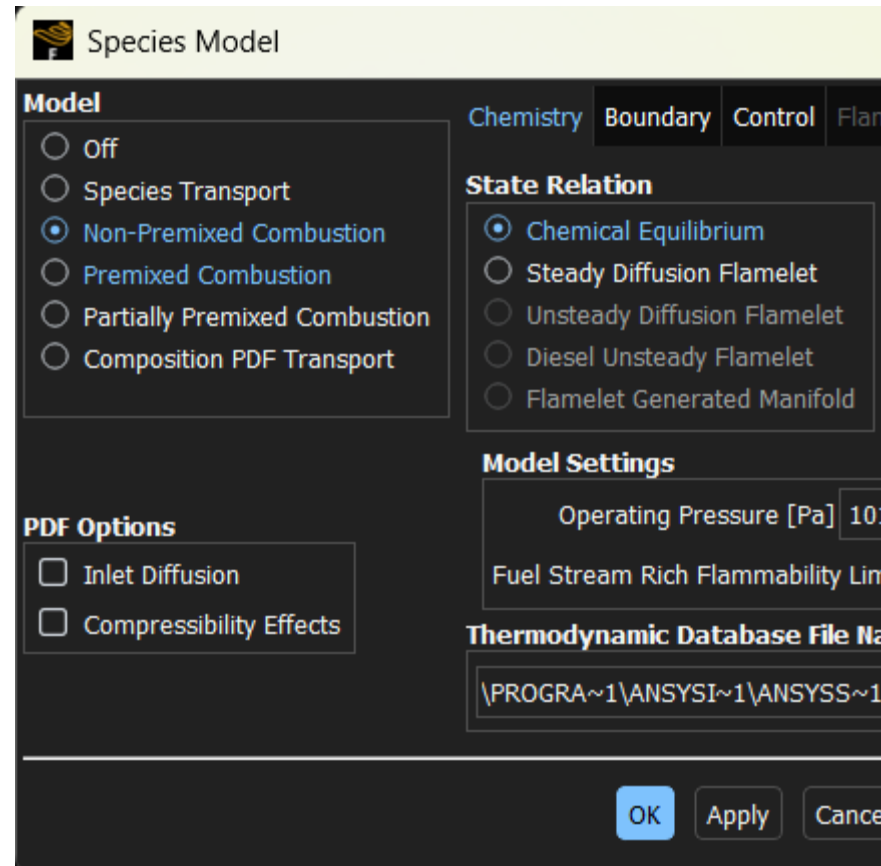
The screenshot shows the 'Mass-Flow Inlet' dialog box with the following settings:

- Zone Name:** air_inlet
- Reference Frame:** Absolute
- Mass Flow Specification Method:** Mass Flow Rate
- Mass Flow Rate [kg/s]:** 0.013562
- Supersonic/Initial Gauge Pressure [Pa]:** 0
- Direction Specification Method:** Normal to Boundary
- Turbulence:**
 - Specification Method:** Intensity and Hydraulic Diameter
 - Turbulent Intensity [%]:** 5
 - Hydraulic Diameter [mm]:** 30

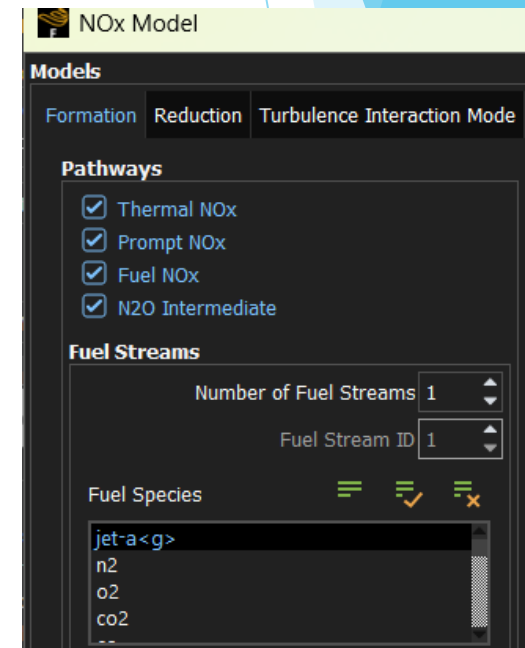
At the bottom, there are three buttons: 'Apply', 'Close', and 'Help'.

Species conditions

- Select the Species model, then **Non-Premixed Combustion**.
- Select **Chemical equilibrium state relation**.
- Then to Boundary tab and give the fuel species accordingly, in our case its only kerosene so jet a<g> = 1.
- Then calculate/generate the PDF Table which resulted in **20 Species**.
- Then the NOx Model and select the corresponding pathways.



Species	Fuel	Oxid
ch4	0	0
h2	0	0
jet-a<g>	1	0
n2	0	0.78992
o2	0	0.21008



Injection Settings

 Discrete Phase Model ✕

Interaction
☒ Interaction with Continuous Phase
☐ Update DPM Sources Every Flow Iteration
DPM Iteration Interval

Particle Treatment
☐ Unsteady Particle Tracking

Contour Plots for DPM Variables
☐ Mean Values

Tracking | Physical Models | UDF | Numerics | Parallel

Tracking Parameters
Max. Number of Steps

☒ Specify Length Scale
Length Scale [mm]

Tracking Option
☐ High-Res Tracking

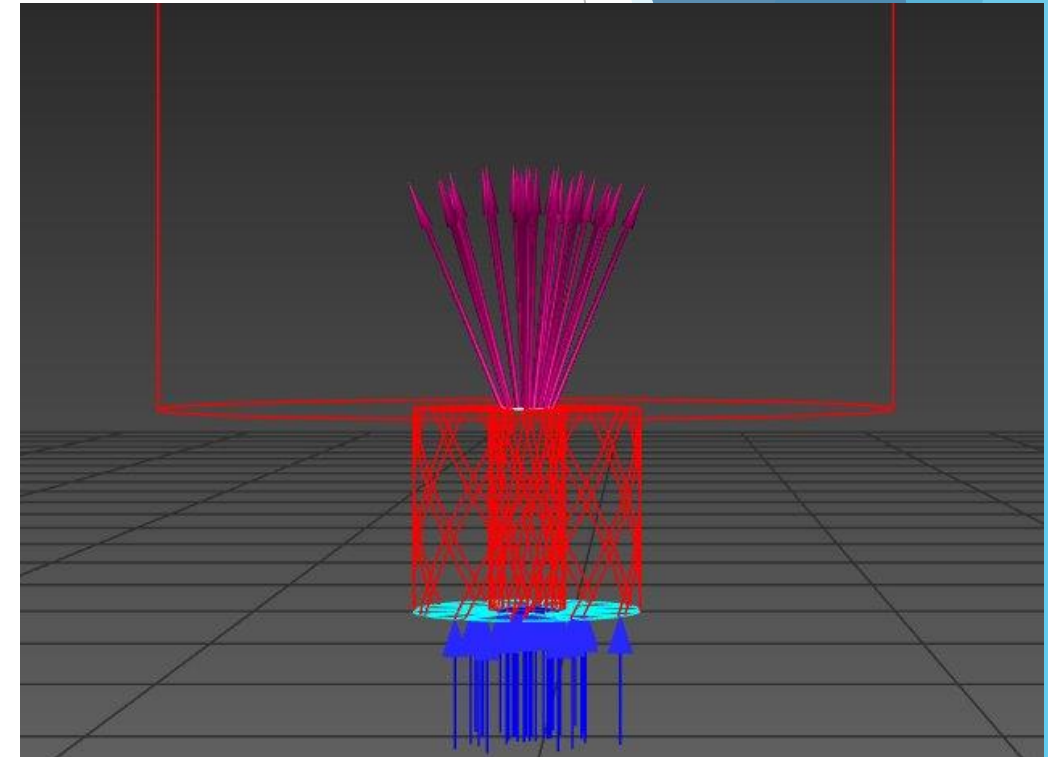
OK

Injections...

DEM Collisions...

Cancel

Help



Set Injection Properties

Injection Name kerosene-injection		Injection Type cone	
Reference Frame global			
Particle Type ☐ Massless ☐ Inert ☒ Droplet ☐ Combusting ☐ Multicomponent		Laws ☐ Custom	
Material kerosene-liquid	Diameter Distribution uniform	Oxidizing Species 	Discrete Phase Domain none
Evaporating Species jet-a<g>	Devolatilizing Species 	Product Species 	
Point Properties	Physical Models	Turbulent Dispersion	Parcel
Wet Combustion			
Components			
UDF			
Multiple Reactions			
Cone Injector Parameters			
Cone Type solid-cone		Uniform Massflow Distribution ☐	
Variable		Value	
X-Position [mm]		0	
Y-Position [mm]		40	
Z-Position [mm]		0	
Diameter [mm]		0.013	
Temperature [K]		300	
Azimuthal Start Angle [deg]		0	
Azimuthal Stop Angle [deg]		360.000000573	
X-Axis		0	
Y-Axis		1	
Z-Axis		0	
Velocity Magnitude [m/s]		30	
Cone Half Angle [deg]		22.5	
Outer Radius [mm]		5	
Total Flow Rate [kg/s]		0.000696	

Stochastic Tracking

Dispersion Model

discrete-random-walk

☐ Random Eddy Lifetime

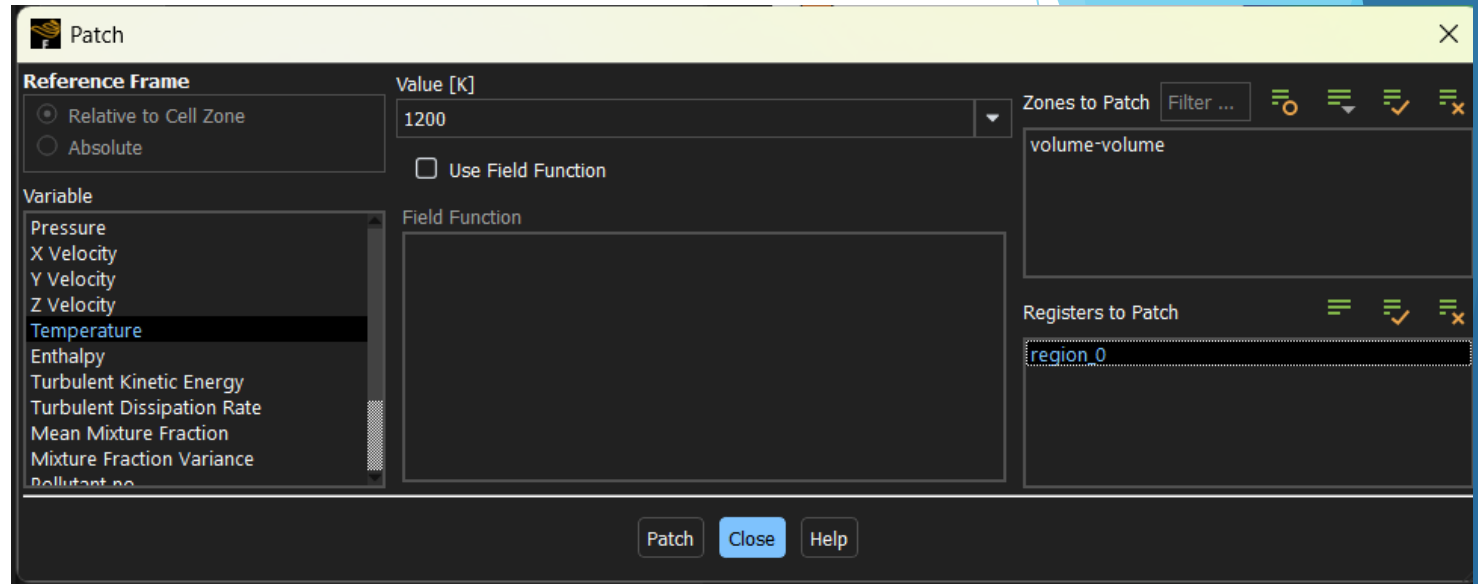
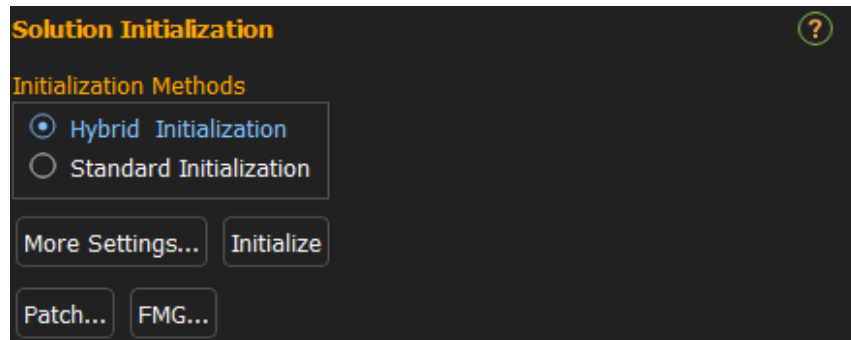
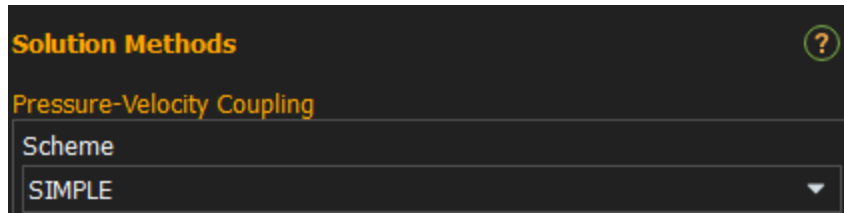
Number of Tries

20

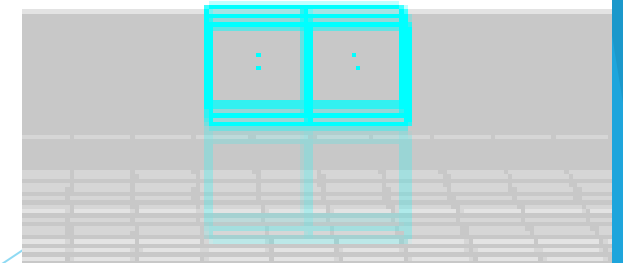
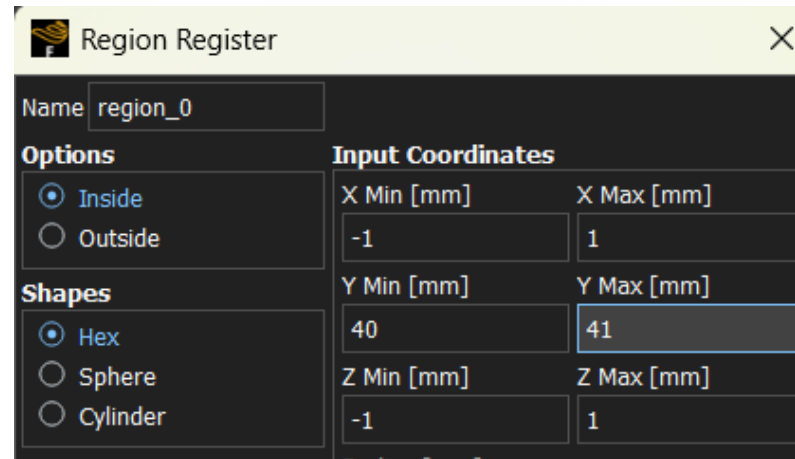
Time Scale Constant

0.15

Initialization and Patch Settings



- Using the " SIMPLE " coupling in solutions ,
Intialise the solution and run the cold flow.
- Once the cold flow is done, give Species &
injection and Boundary conditions as
mentioned in above slides.
- Patch the solution with a proper region , which
in our case is a small region register ,
consisting of 4 cells .
- Now run the solution until it converges, run
iterations with an interval of 5000 .
- Convergence can be viewed in solution status.



Solver settings

Equations	
Flow	True
Turbulence	True
Pollutant no	True
Pollutant hcn	True
Pollutant nh3	True
Pollutant n2o	True
Energy	True
Pdf	True
Numerics	
Absolute Velocity Formulation	False
Under-Relaxation Factors	
Pressure	0.3

Density	1
Body Forces	1
Momentum	0.7
Turbulent Kinetic Energy	0.8
Turbulent Dissipation Rate	0.8
Turbulent Viscosity	1
Pollutant no	1
Pollutant hcn	1
Pollutant nh3	1
Pollutant n2o	1
Energy	1
Temperature	1
Mean Mixture Fraction	1

Mixture Fraction Variance	0.9
Discrete Phase Sources	0.5
- Pressure-Velocity Coupling	
Type	SIMPLE
- Discretization Scheme	
Pressure	Second Order
Momentum	Second Order Upwind
Turbulent Kinetic Energy	First Order Upwind
Turbulent Dissipation Rate	First Order Upwind
Pollutant no	Second Order Upwind
Pollutant hcn	Second Order Upwind
Pollutant nh3	Second Order Upwind
Pollutant n2o	Second Order Upwind

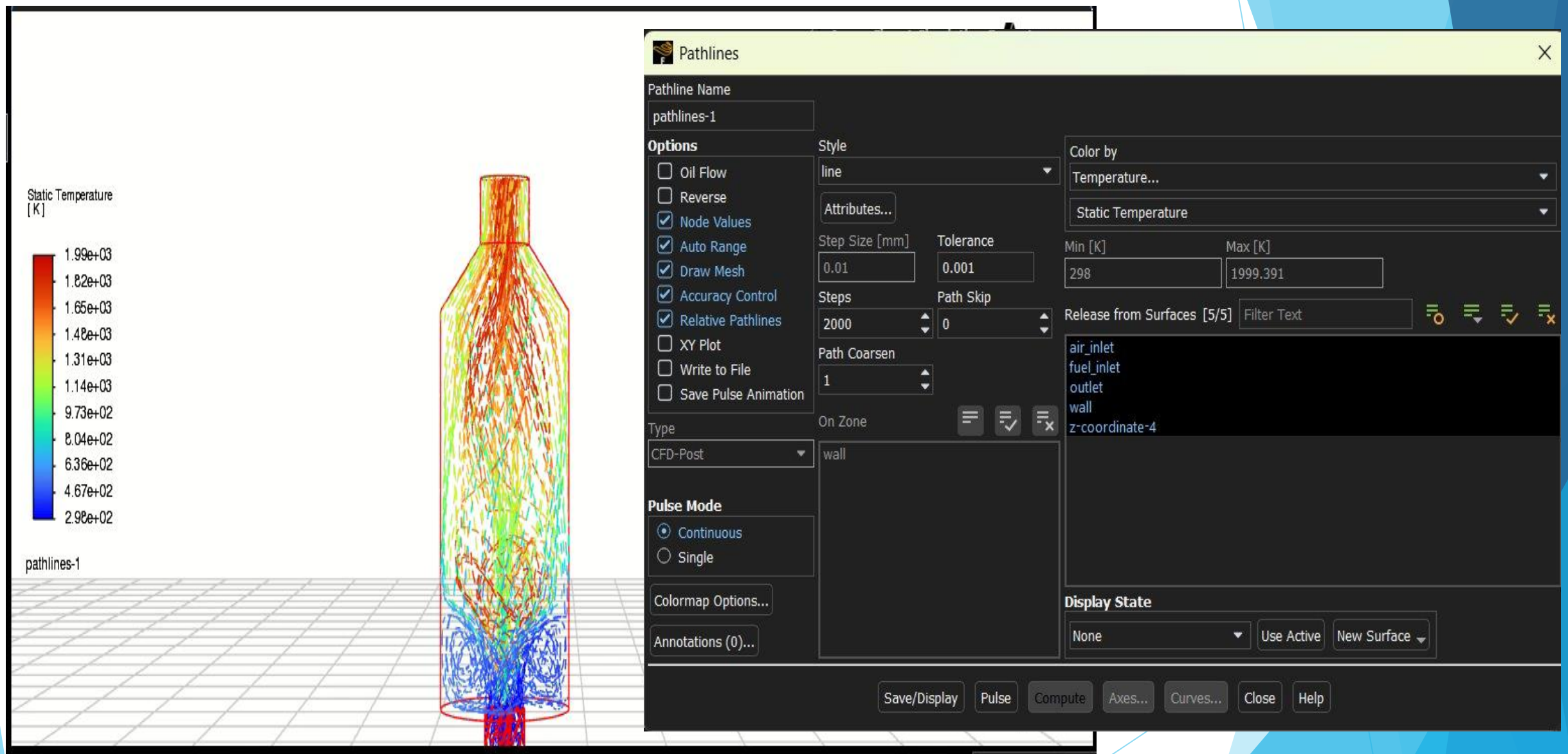
Energy	Second Order Upwind
Mean Mixture Fraction	Second Order Upwind
Mixture Fraction Variance	Second Order Upwind
- Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1
Maximum Static Temperature [K]	5000
Minimum Turb. Kinetic Energy [m ² /s ²]	1e-14
Minimum Turb. Dissipation Rate [m ² /s ³]	1e-20
Maximum Turb. Viscosity Ratio	100000

Results

Solution Status

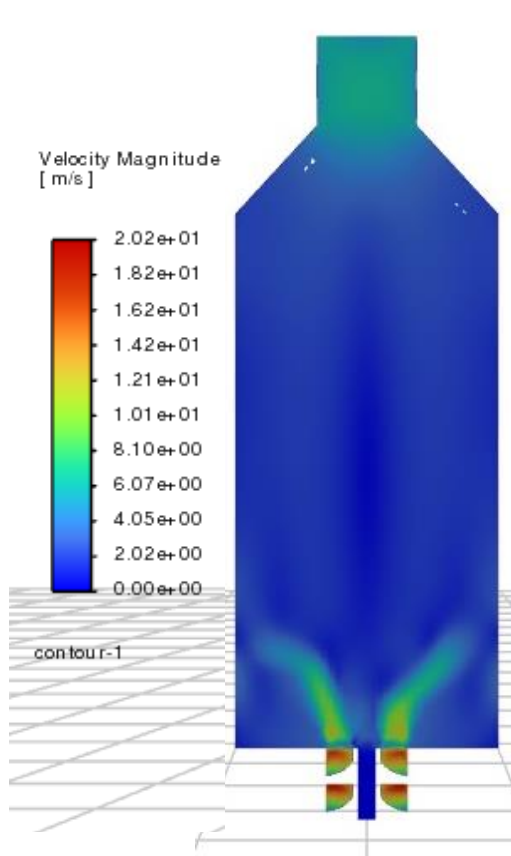
Iterations: 42835 (Equivalence ratio - 1.2 case)

	Value	Absolute Criteria	Convergence Status
continuity	0.008562584	0.001	Not Converged
x-velocity	0.000095123	0.0001	Converged
y-velocity	0.000092232	0.0001	Converged
z-velocity	0.000091345	0.0001	Converged
energy	0.000042374	0.0001	Converged
k	0.0005850106	0.001	Converged
epsilon	0.0000077848	1e-05	Converged
fmean	0.00003354853	0.0001	Converged
fvar	0.00006802159	0.0001	Converged
pollut_no	0.00042911	1e-05	Not Converged

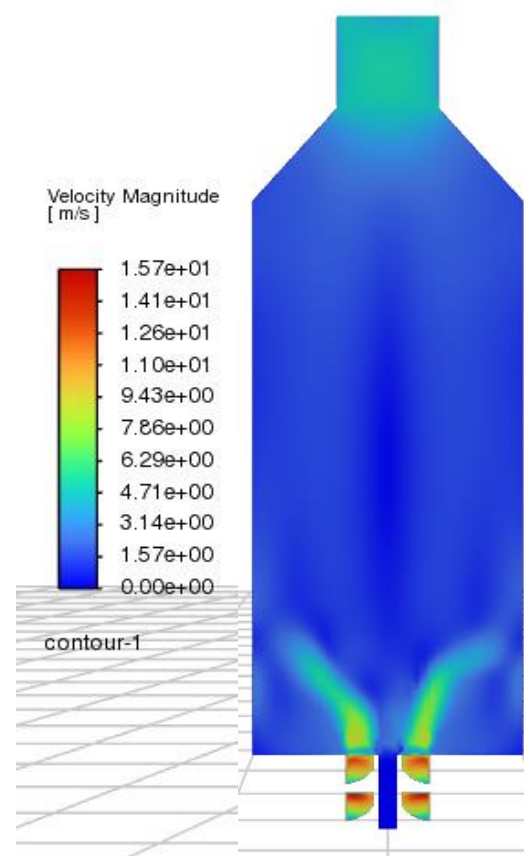


Pathlines of temperature post combustion in a continuous pulse mode

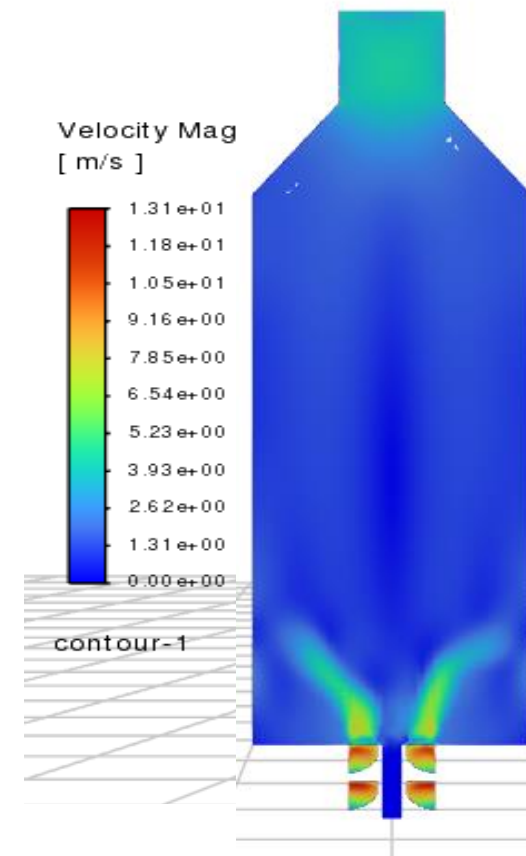
Axial velocity contours for cold flow



Equivalence ratio(ϕ) – 0.8



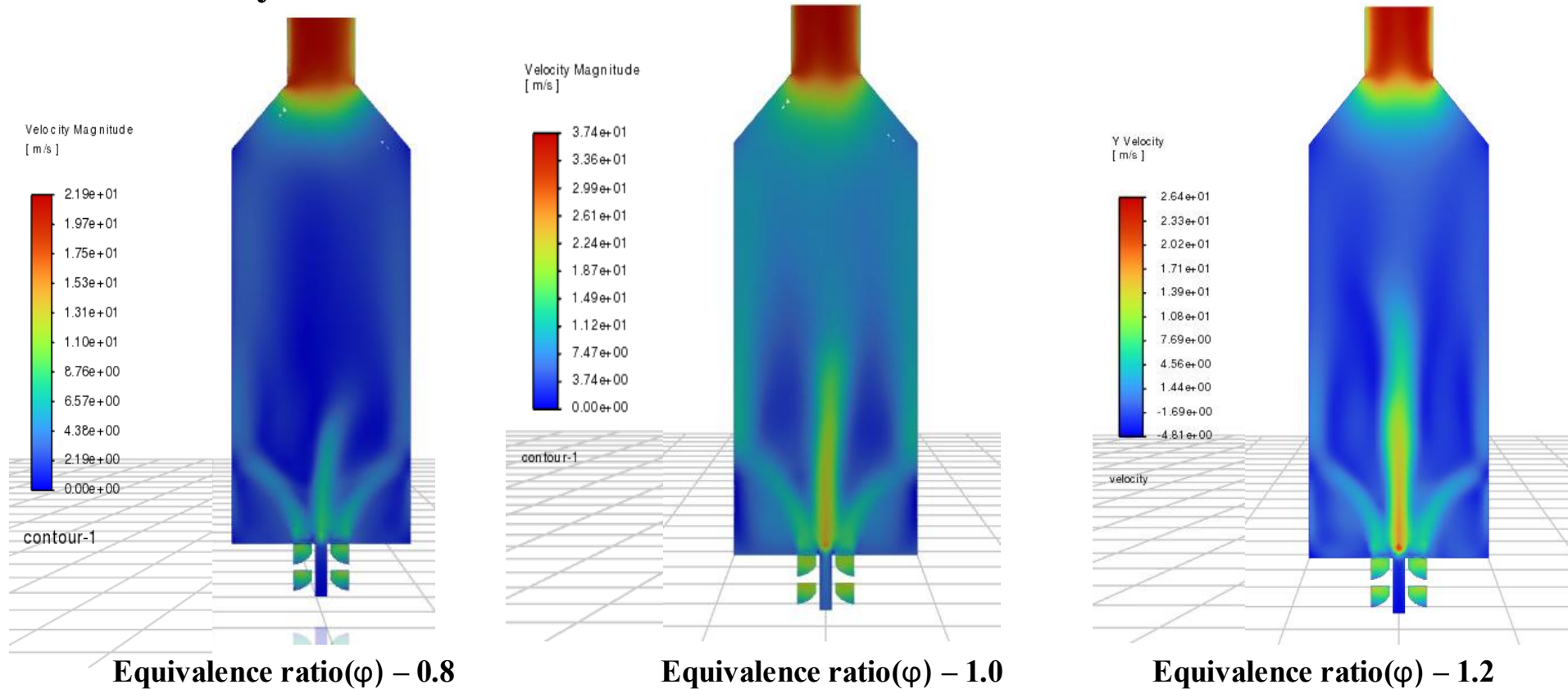
Equivalence ratio(ϕ) – 1.0



Equivalence ratio(ϕ) – 1.2

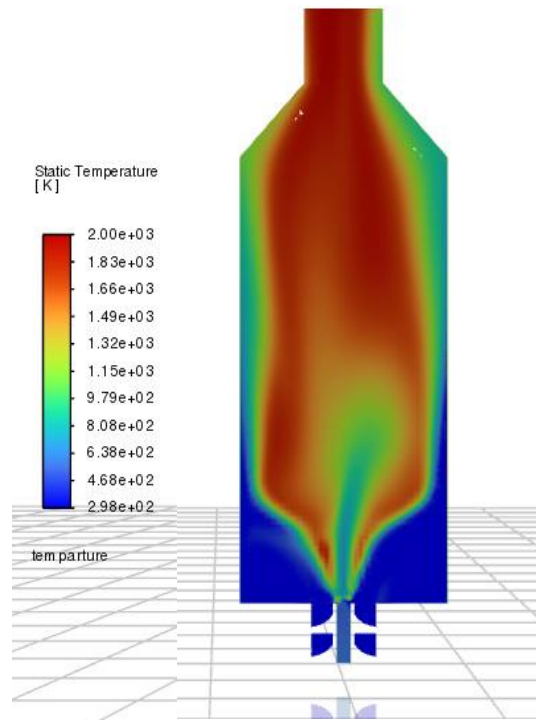
- Velocity magnitude decreases from left to right, indicating that higher mass flow produces higher inlet jet velocity and stronger flow development.
- Jet penetration and spreading reduce as velocity decreases, leading to weaker mixing and a shorter core flow region in lower velocity cases.

Axial velocity contours – After combustion

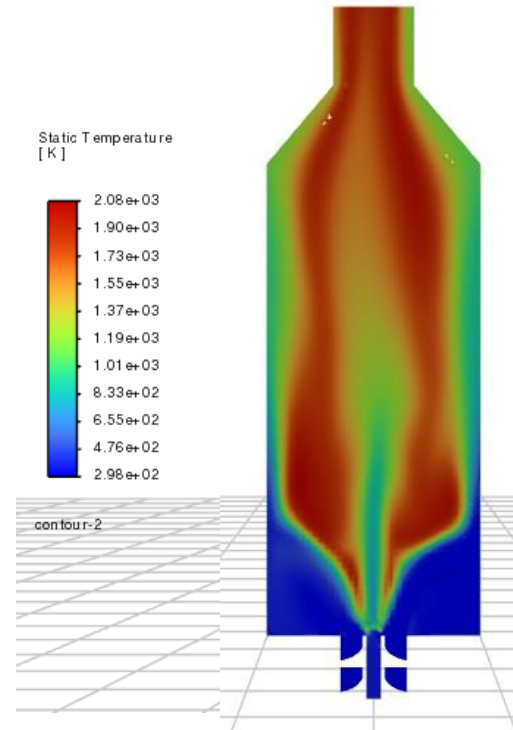


- Combustion causes heat release $\rightarrow$ temperature rise $\rightarrow$ gas expansion $\rightarrow$ density decrease, which accelerates the flow. This effect is strongest at $\phi = 1$ (stoichiometric) because heat release is maximum, giving the highest acceleration and stronger axial jet.
- Velocity follows: $\phi = 0.8 < \phi = 1.2 < \phi = 1.0$.
- Maximum at $\phi = 1$ (maximum heat release), Lower at lean (0.8) due to heat dilution, Lower at rich (1.2) due to incomplete combustion. Thus, Velocity $\propto$ Temperature $\propto$ Heat release, peaking near $\phi \approx 1$.

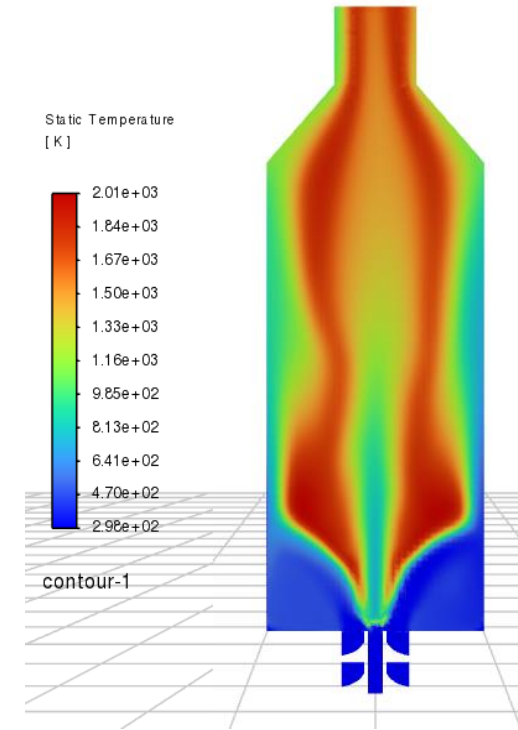
Temperature contour



Equivalence ratio(ϕ) – 0.8



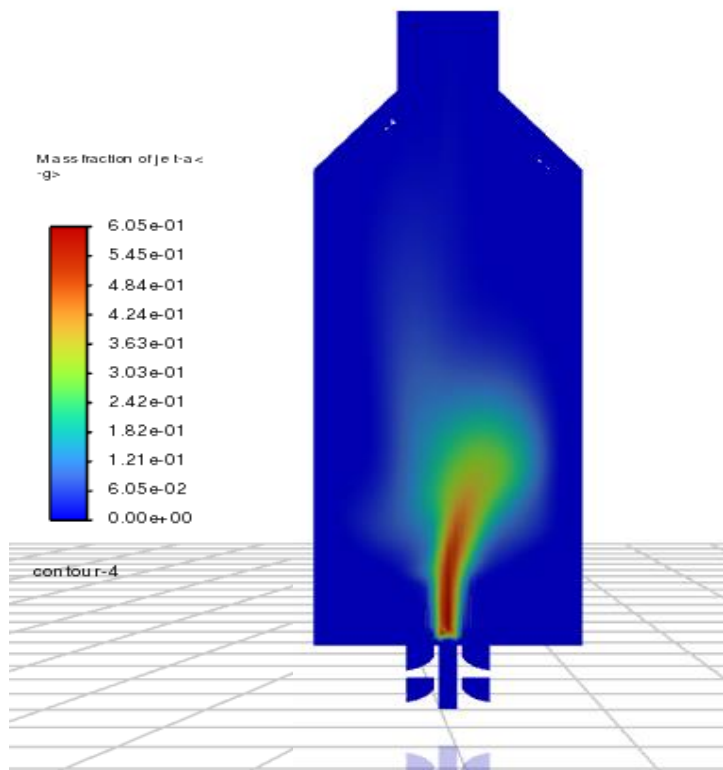
Equivalence ratio(ϕ) – 1.0



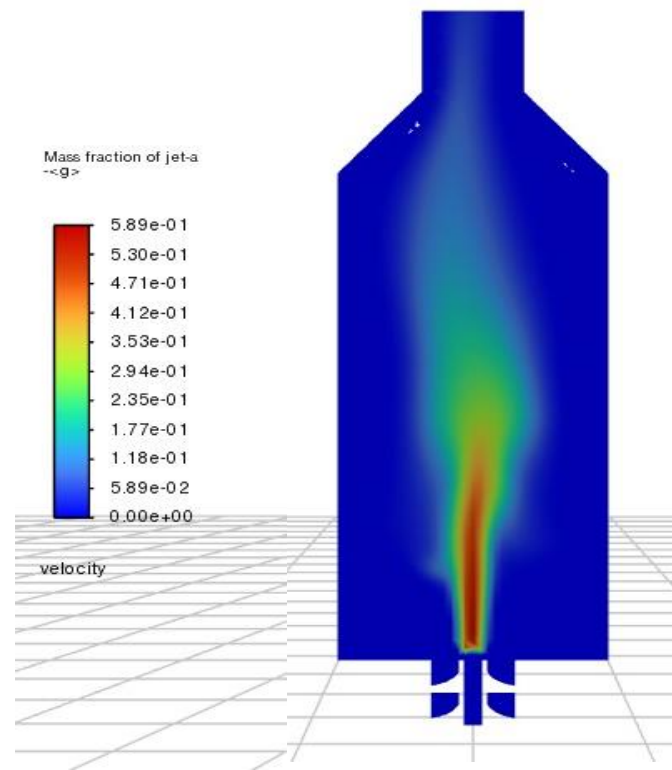
Equivalence ratio(ϕ) – 1.2

- The high-temperature region forms an elongated flame core because heat is released where fuel–air mixing and reactions occur . The flame anchors near the injector and extends upward due to jet momentum and convection, while cooler regions near walls indicate heat losses and excess air dilution.
- **Temperature follows:** $\phi = 0.8 < \phi = 1.2 < \phi = 1.0$
- Maximum at $\phi = 1$ (**stoichiometric**) due to optimal combustion, Lower at **lean (0.8)** due to excess air cooling, Lower at **rich (1.2)** due to incomplete combustion . Thus, **Temperature $\propto$ heat release**, peaking near $\phi \approx 1$.

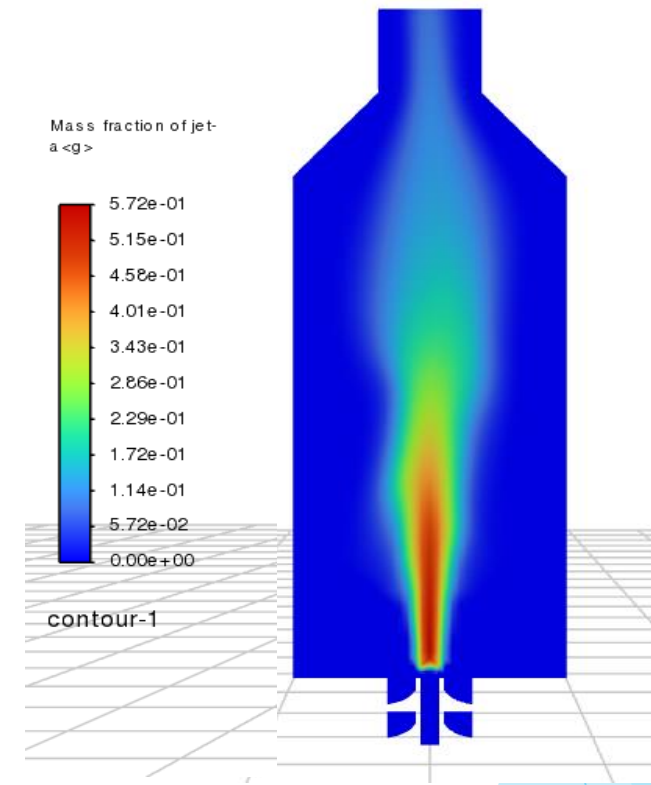
Contour – Mass Fraction of Fuel (kerosene-Jet –a<g>)



Equivalence ratio(ϕ) – 0.8



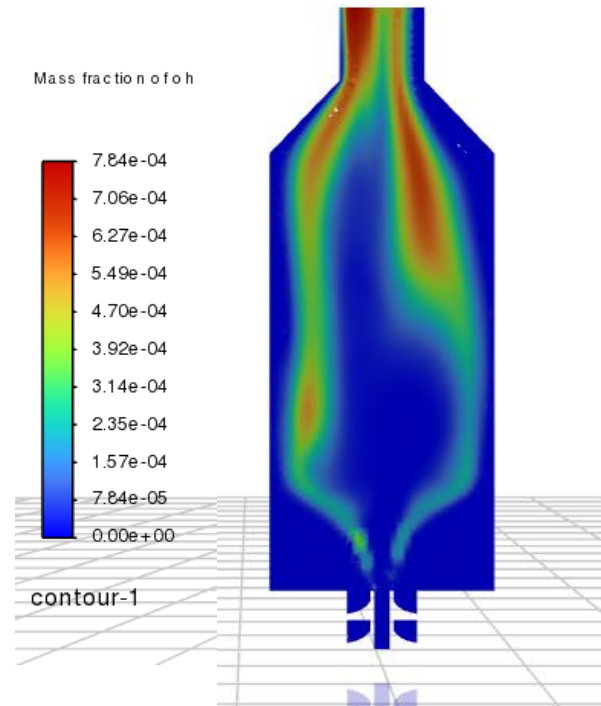
Equivalence ratio(ϕ) – 1.0



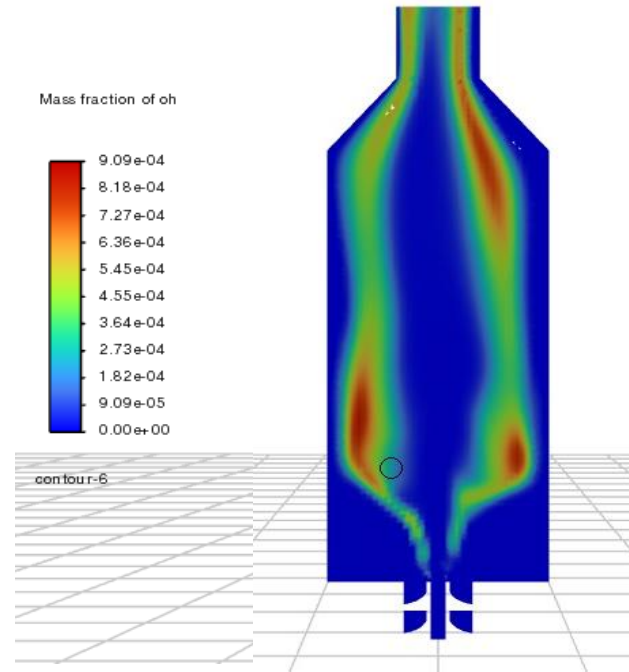
Equivalence ratio(ϕ) – 1.2

- **Fuel penetration increases with equivalence ratio:** The jet becomes longer and more intense from $\phi = 0.8 \rightarrow 1 \rightarrow 1.2$, indicating higher fuel concentration and stronger core for richer mixtures.
- **Mixing is better in lean case:** At $\phi = 0.8$, the fuel spreads wider and diffuses faster (more blue/green region), while at $\phi = 1.2$, the jet is narrow and concentrated $\rightarrow$ poorer mixing and higher unburnt fuel core.

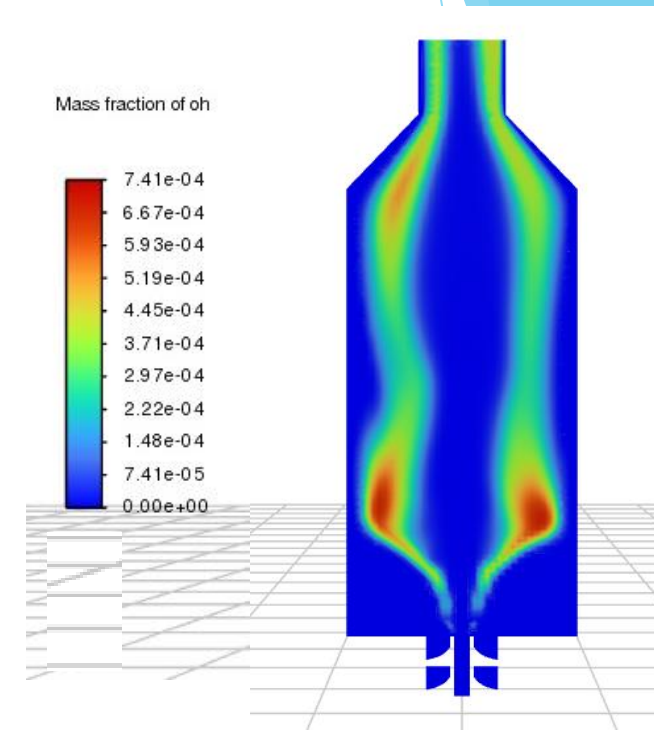
Contour – Mass Fraction of OH



Equivalence ratio(ϕ) – 0.8



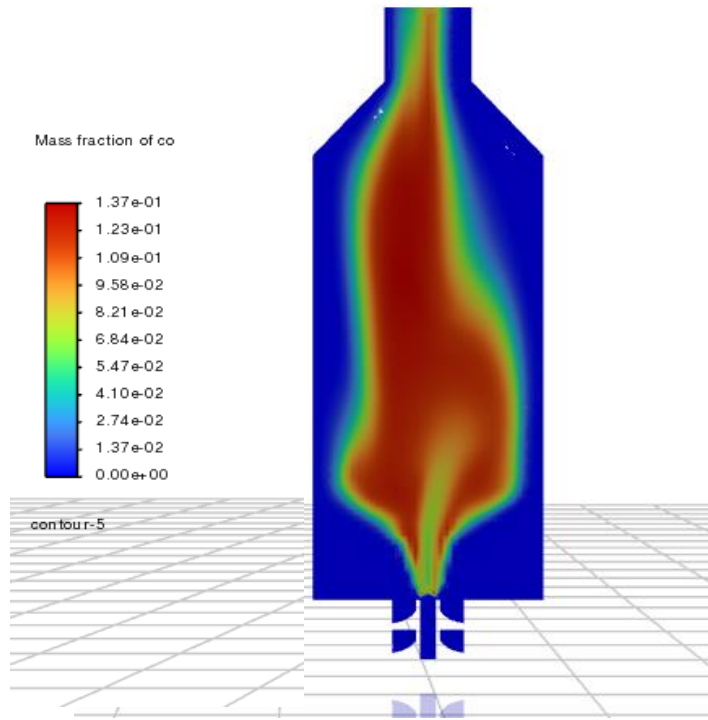
Equivalence ratio(ϕ) – 1.0



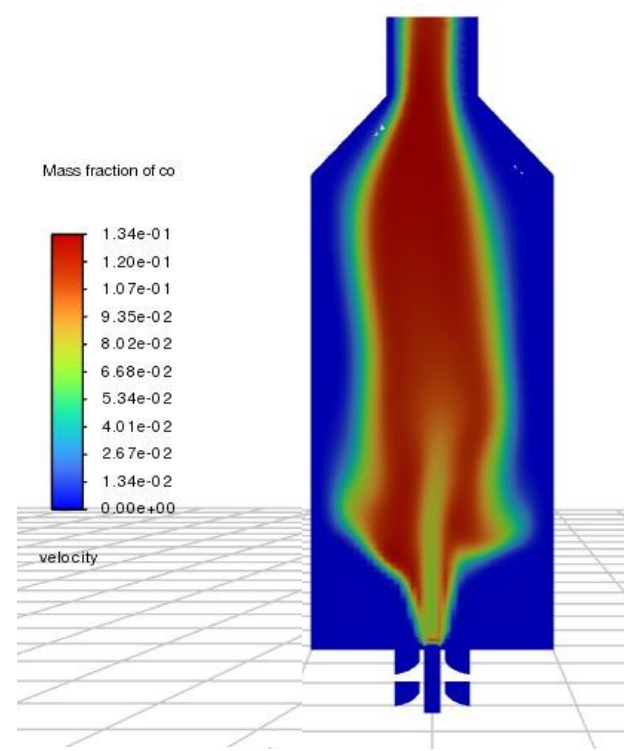
Equivalence ratio(ϕ) – 1.2

- The OH contours form a flame-shaped/elongated region along the shear layers because OH is a marker of high-temperature reaction zones. It appears where fuel and oxidizer mix effectively.
- The upward stretching of OH indicates jet-driven transport, while the widening shows turbulent mixing and flame spreading.
- Trend & relationship with equivalence ratio - **OH intensity follows: $\phi = 0.8 < \phi = 1.2 < \phi = 1.0$.**
- Maximum at $\phi = 1$ (strongest reaction rate), Lower at lean (0.8) due to temperature dilution Lower at rich (1.2) due to lack of oxygen.

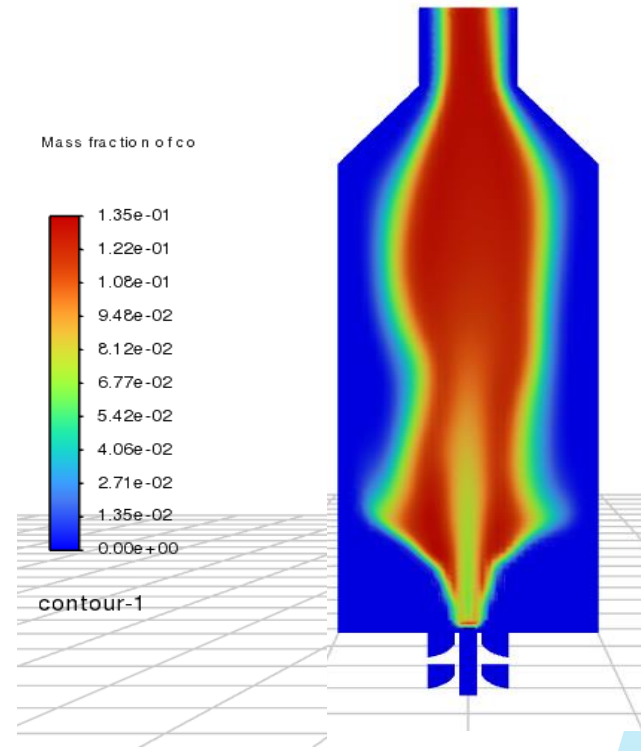
Contour – Mass Fraction of CO



Equivalence ratio(ϕ) – 0.8



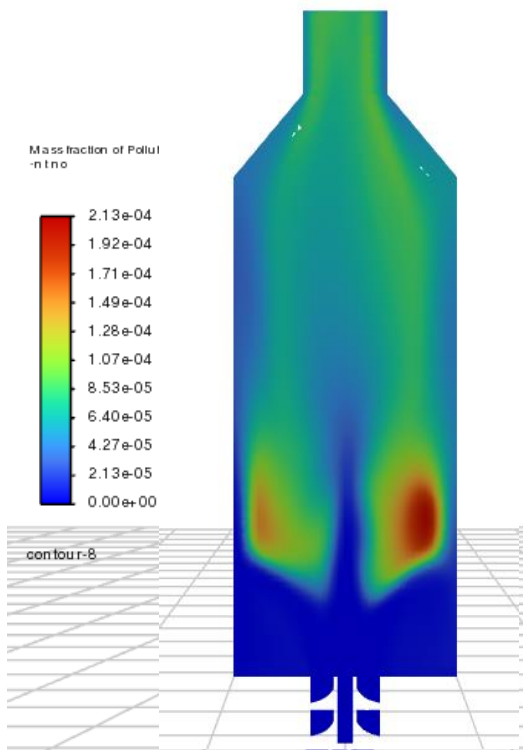
Equivalence ratio(ϕ) – 1.0



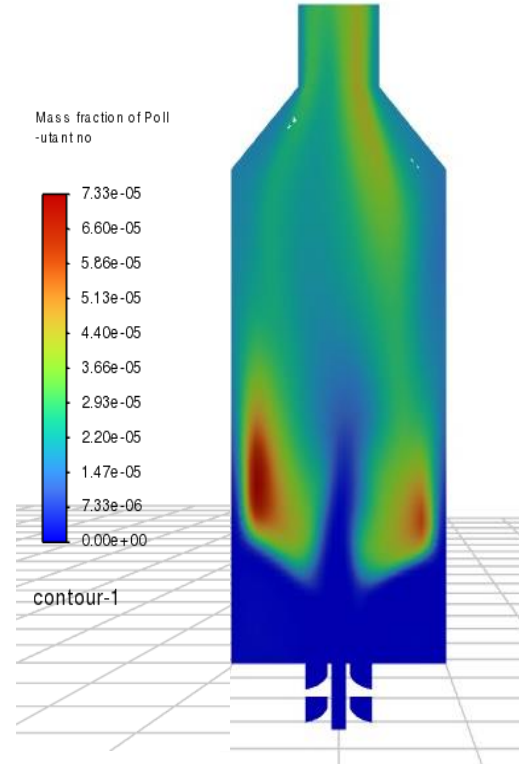
Equivalence ratio(ϕ) – 1.2

- CO forms in fuel-rich and incomplete combustion zones, mainly along the central core of the flame where oxygen is limited.
- As gases move upward, CO persists because oxidation to CO₂ requires sufficient O₂ and residence time, so high CO regions appear elongated and concentrated in the core.
- CO follows: $\phi = 0.8 < \phi = 1.0 < \phi = 1.2$.
- Lowest at lean (0.8) due to excess oxygen and complete combustion, Moderate at stoichiometric (1.0), Maximum at rich (1.2) due to oxygen deficiency. Thus, $\text{CO} \propto \text{fuel richness} \propto (\text{lack of O}_2)$, increasing with equivalence ratio.

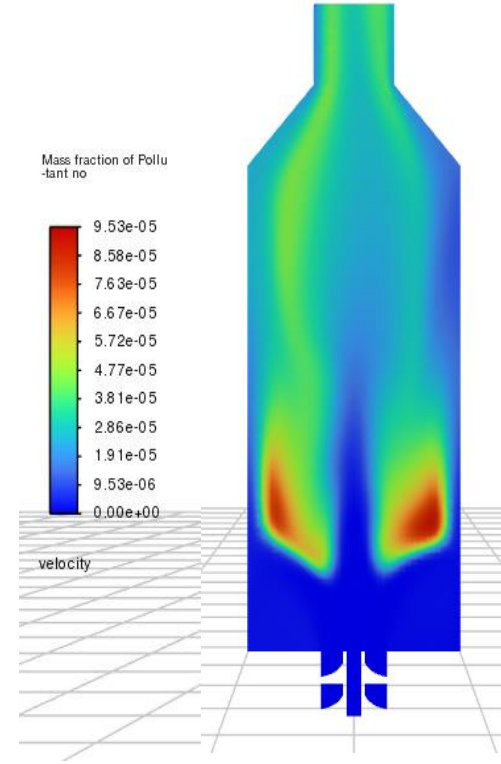
Contour – Mass Fraction of NO



Equivalence ratio(ϕ) – 0.8



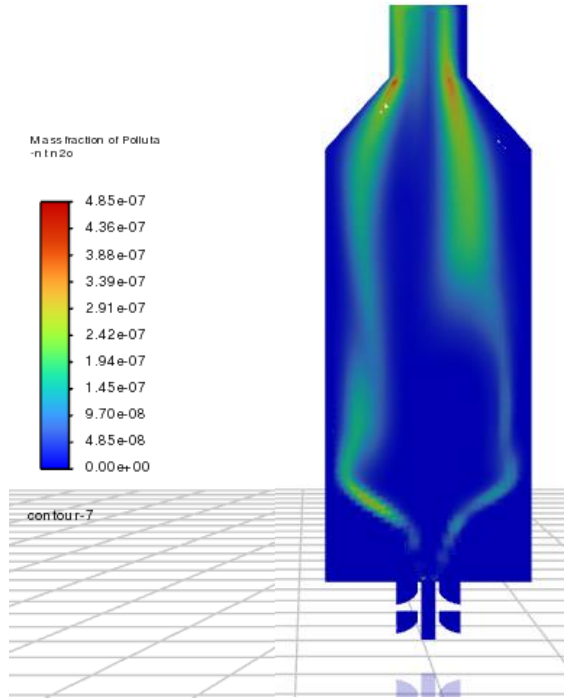
Equivalence ratio(ϕ) – 1.0



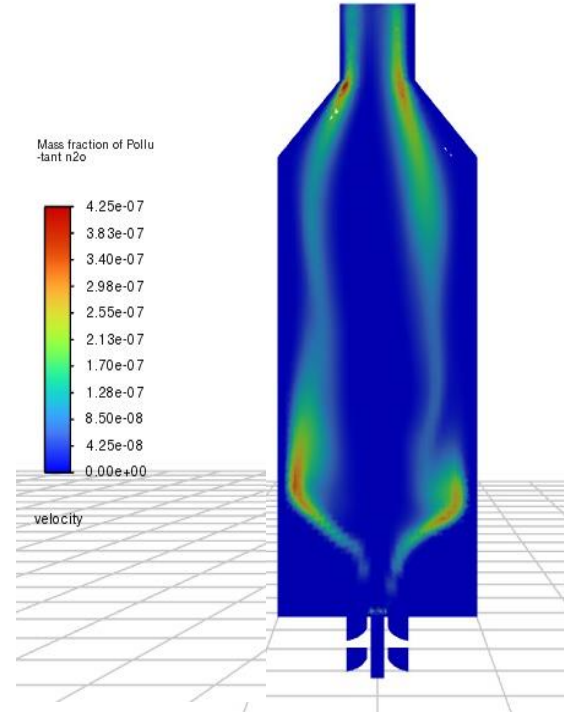
Equivalence ratio(ϕ) – 1.2

- NO forms mainly through the PDF mechanism, which is highly sensitive to high temperature and presence of O_2 . Hence, NO contours follow the reaction zones and appear slightly downstream where gases have sufficient residence time at high temperature.
- Trend & relationship with equivalence ratio - **NO follows: $\phi = 0.8 > \phi = 1.0 > \phi = 1.2$.**
- Highest at lean (0.8) due to excess O_2 + high temperature, Moderate at stoichiometric (1.0), Lowest at rich (1.2) due to oxygen deficiency .
- Thus, $NO \propto (\text{Temperature} \times O_2 \text{ availability})$ and peaks on the lean side of combustion.

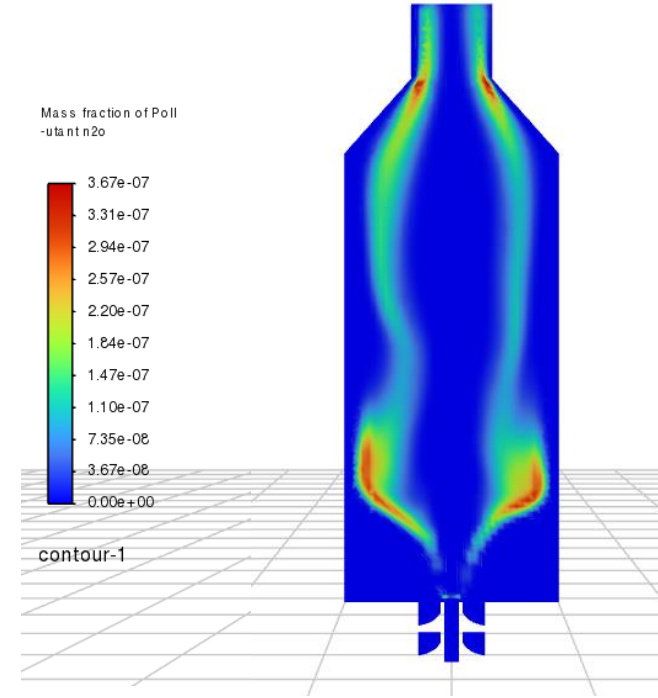
Contour – Mass Fraction of N₂O.



Equivalence ratio(ϕ) – 0.8



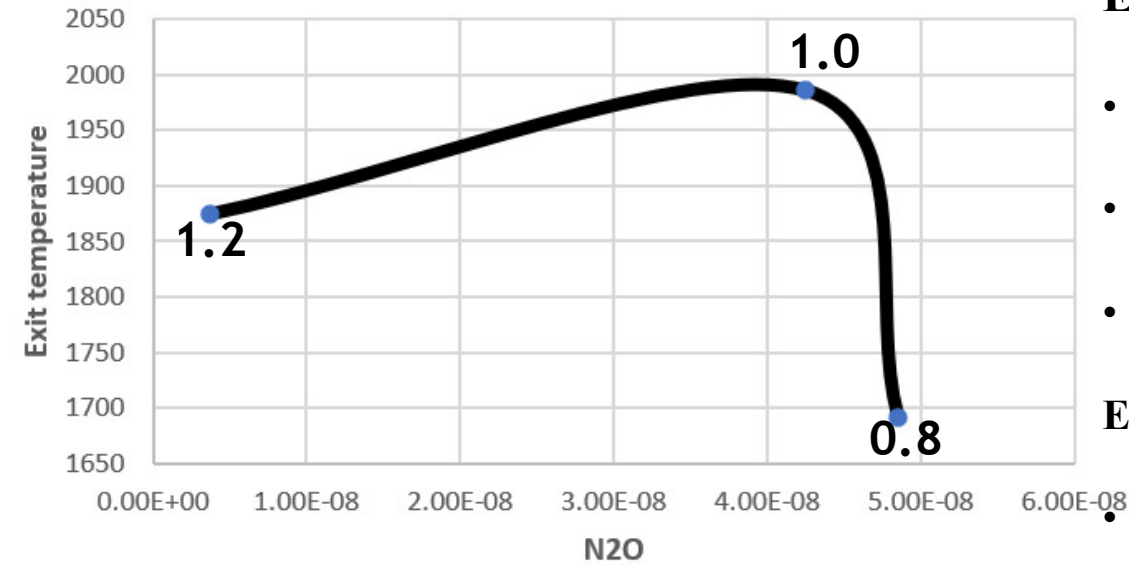
Equivalence ratio(ϕ) – 1.0



Equivalence ratio(ϕ) – 1.2

- N₂O forms mainly through intermediate (prompt NO_x) pathways in moderate-temperature, fuel-rich or partially premixed zones, not at the hottest core like NO. Hence, it appears in the outer shear layers and near recirculation zones, where mixing is incomplete and temperatures are not extremely high.
- Trend & relationship with equivalence ratio - N₂O follows : $\phi = 1.2 > \phi = 1.0 > \phi = 0.8$.
- Highest in rich (1.2) due to fuel-rich chemistry and intermediate reactions, Moderate at stoichiometric (1.0), Lowest in lean (0.8) due to complete oxidation and lower intermediate species.
- N₂O $\propto$ fuel-rich intermediate reactions, increasing with equivalence ratio (opposite trend to NO).

Exit temperature vs N2O



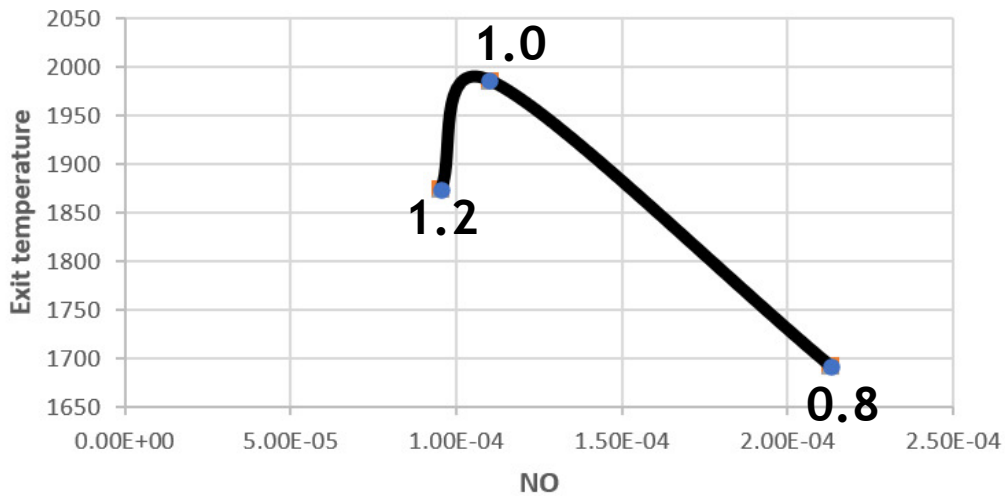
Exit Temperature vs N₂O.

- N₂O increases as temperature decreases from (lean to rich shift).
- N₂O is stable in moderate/low-temperature zones and decomposes at high temperatures.
- N₂O follows : $\phi = 1.2 > \phi = 1.0 > \phi = 0.8$.

Exit Temperature vs NO

- NO peaks at highest temperature ($\phi = 1$): NO formation is strongly dependent on high temperature (thermal NO mechanism), hence maximum at stoichiometric condition.
- NO decreases for lean and rich cases – it follows - $\phi = 0.8 > \phi = 1.0 > \phi = 1.2$.

Exit temperature vs NO



Why Area-Weighted Average at Outlet?

- Represents true outlet conditions across the entire exit area
- Accounts for spatial non-uniformity in flow properties
- Provides reliable bulk values for performance and emission comparison

Area	[m^2]
z-coordinate-4	0.052380362
Area-Weighted Average Static Temperature	[K]
outlet	1692.3973
Area-Weighted Average Mass fraction of Pollutant no	
outlet	2.13e-04
Area-Weighted Average Mass fraction of Pollutant n2o	
outlet	4.85e-08

Area	[m^2]
z-coordinate-4	0.052380362
Area-Weighted Average Static Temperature	[K]
outlet	1986.7564
Area-Weighted Average Mass fraction of Pollutant no	
outlet	1.10e-04
Area-Weighted Average Mass fraction of Pollutant n2o	
outlet	4.25e-08

Area	[m^2]
z-coordinate-4	0.052380362
Area-Weighted Average Static Temperature	[K]
outlet	1874.6564
Area-Weighted Average Mass fraction of Pollutant no	
outlet	9.53e-05
Area-Weighted Average Mass fraction of Pollutant n2o	
outlet	3.67e-09